The optimal two-colour six-point constant

for Besicovitch's 1/2 problem

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Abstract

Besicovitch's 1/2 conjecture asserts that a planar Borel set of finite $\mathcal{H}^1$ measure whose lower density exceeds 1/2 almost everywhere is countably 1-rectifiable. Following the linear-programming strategy introduced by De Lellis, Glaudo, Massaccesi and Vittone, the conjecture is attacked through a family of finite packing problems whose optimal constants bound the true density threshold from above. We solve the two-colour, six-centre instance of that family exactly. Its optimal constant is the algebraic number

$$\theta_6 = s_* = 0.69330642182590487269067841440371095 1\ldots,$$

specified by an isolated pair of radical equations rather than by a decimal, and the extremal configuration is unique up to the evident symmetries. Combined with a direct six-centre transfer to the Besicovitch pair condition and the standard rectifiability machinery, this gives the bound $\sigma_1(\mathbb{R}^2) \leq s_*$ for the planar rectifiability threshold. The proof is computer-assisted only in explicitly isolated geometric inequalities: every certificate is a finite, outward-rounded interval computation over rational data, and no floating-point sign, hash, or exit code is a proof object. A companion Lean 4 development formalizes the argument; its present status is described in Section 12.

Contents

1 Introduction

1.1 The problem

Let $E \subset \mathbb{R}^2$ be a Borel set with $\mathcal{H}^1(E) < \infty$, where $\mathcal{H}^1$ denotes 1-dimensional Hausdorff measure. The *lower density* of E at a point x is

$$\Theta^1_*(E, x) = \liminf_{r \downarrow 0} \frac{\mathcal{H}^1(E \cap B(x, r))}{2r},$$

the normalization $2r$ being the diameter of the ball, so that a straight segment has density 1 at its interior points. The set E is *countably* 1*-rectifiable* if $\mathcal{H}^1$-almost all of it can be covered by countably many Lipschitz curves.

A classical theorem of Besicovitch says that a set with lower density strictly larger than 3/4 almost everywhere is rectifiable, and he conjectured that 3/4 can be replaced by 1/2.

Conjecture 1 (Besicovitch's 1/2 conjecture)**.** If $E \subset \mathbb{R}^2$ is Borel with $\mathcal{H}^1(E) < \infty$ and $\Theta^1_*(E, x) > 1/2$ for $\mathcal{H}^1$-a.e. $x \in E$, then E is countably 1-rectifiable.

The threshold 1/2 is the natural one: it is the value attained at the endpoint of a segment, and no known construction of a purely 1-unrectifiable set of finite length achieves a lower density above it.

It is convenient to name the optimal constant. Write

$$\sigma_1(\mathbb{R}^2) = \inf\{\beta \geq 0 : \text{ every finite-measure Borel set with } \Theta^1_* \geq \beta \text{ a.e. is countably 1-rectifiable}\}.$$

Besicovitch's conjecture is the assertion $\sigma_1(\mathbb{R}^2) = 1/2$.

1.2 History

The recorded progress on the upper bound for $\sigma_1(\mathbb{R}^2)$ is short and hard-won.

year	bound	source
1928	$1 - 10^{-2576}$	Besicovitch
1938	$3/4$	Besicovitch
1992	$(2 + \sqrt{46})/12 = 0.73186\ldots$	Preiss and Tišer
2024	0.7	De Lellis, Glaudo, Massaccesi and Vittone

The 1992 bound of Preiss and Tišer stood for more than three decades. Its method rests on a two-point inequality for straight measures, and the sharp threshold obtainable from that method is the unique positive root of $8s^3 + 4s^2 - 3s - 3$, namely $0.72655\ldots$.

The decisive structural advance is due to De Lellis, Glaudo, Massaccesi and Vittone, who recast the problem as a family of finite-dimensional optimization problems. Their insight is that the passage from a density hypothesis to rectifiability can be routed through a purely finite statement about disjoint balls centred at finitely many points, and that this finite statement is a linear program once the centres are fixed. The optimization approach in the present work, and in the accompanying Lean development, is directly inspired by theirs.

1.3 What is proved here

The finite problems in this family are indexed by the number of centres and by how those centres are grouped. The instance solved here uses two *colours* — corresponding to the two sets E_1, E_2 that a separation argument produces — and three centres of each colour: a root and two children. Six centres in all. Write θ_6 for the optimal constant of that instance; a precise definition is given in Section 4.

Theorem 1.1 (Main theorem; Theorem 9.1)**.** *The optimal two-colour six-point constant is the algebraic number*

$$\theta_6 = s_* = 0.69330642182590487269067841440371095 1\ldots,$$

described exactly in Section 5. The normalized extremal configuration is unique up to reflection in the root line, interchange of the colours, and relabelling of the two children within either colour.

The constant is not a numerical artefact of an optimizer. It is pinned by two radical equations inside an explicit rational isolation box, and the decimal expansion above appears nowhere in the proof. What makes it that particular number is that three distinct packing mechanisms tie there simultaneously; this is explained in Section 3.

Theorem 1.2 (Transfer; Section 11)**.** $\sigma_1(\mathbb{R}^2) \le s_*$.

Theorem 1.2 follows from Theorem 1.1 by a direct six-centre argument establishing the Besicovitch pair condition at every parameter $\beta > s_*$, followed by the passage from that condition to rectifiability and an infimum. Both steps are described in Section 11 and are fully formalized.

1.4 Where a computer is used

Three ingredients of the proof are finite interval computations:

1. the isolation of the algebraic endpoint and its associated weights;
2. a list of strict routing separators that prune the case tree;
3. the two analytic inequalities of Section 8, namely a self inequality on one sibling pair and a contraction inequality relating a mixed pair to the two self terms.

In every case the statement being certified is a *universal inequality on an explicitly covered compact domain*, not the output of an optimizer. All interval endpoints are rational or dyadic, every arithmetic operation is rounded outward using integers, and every sign decision is an integer comparison. No random sampling, floating-point comparison, `native_decide`, or unchecked external computation enters a proof step. Section 10 describes the arithmetic, the charts, and the exact domains covered.

1.5 Organization

Sections 2 and 3 are, respectively, a roadmap of the argument and an informal account of the geometry behind it; a reader who wants to know what the proof does, and why the answer is what it is, should be able to stop after them. The details begin in Section 4. Section 5 constructs the exact endpoint, Section 6 proves the lower obstruction, Sections 7 and 8 prove the converse, Section 9 assembles Theorem 1.1, Section 10 documents the certificates, Section 11 deduces Theorem 1.2, and Section 12 reports on the formalization. Three appendices collect the routing algebra, the rational tangent lemma, and the exact weights.

2 Outline of the proof

The whole argument is a chain of five reductions. None of them is long to state; the work is in the last two.

Step 1: from density to a straight measure and a separated pair. Suppose E is a counterexample. Standard measure-theoretic surgery — decomposition into rectifiable and purely unrectifiable parts, differentiation, and rescaling — replaces E by a *straight* measure μ, one for which every measurable set A satisfies $\mu(A) \leq \operatorname{diam} A$, still with lower density above the threshold almost everywhere. Inside such a measure one selects two measurable sets E_1, E_2 at small positive distance, and asks whether some open set meeting both must carry a definite amount of mass *outside* $E_1 \cup E_2$. That question is the Besicovitch pair condition; if it has a positive answer at a parameter β, then β forces rectifiability.

Step 2: from the pair condition to a finite packing problem. Pick approximate closest points of E_1 and E_2 and normalize their distance to 1. Straightness converts the mass in the unit ball around each of these two points into a *sibling pair*: two further points of the same set, at mutual distance at least $2s$, each within distance 1 of its own root. This is exactly six points, in two colours of three. If for every such six-point configuration one can exhibit a family of balls whose total radius is at least $1/(2s)$ times the diameter of their union, then inclusion–exclusion against straightness produces the required leaking open set, and the pair condition holds for every $\beta > s$.

Step 3: the finite problem, and its optimal constant. The finite problem is therefore: for which s does every admissible six-point configuration admit a *packing move* — a choice of at least one centre of each colour and radii, with same-colour balls disjoint — of nonnegative *score*

$$\mathrm{score}_s = \sum_i r_i - \frac{D}{2s}, \qquad D = \operatorname{diam}\Big(\bigcup_i \bar{B}(x_i, r_i)\Big)?$$

The optimal constant θ_6 is the infimum of such s. Since the transfer of Step 2 needs only $\beta > s$, the value θ_6 itself is what bounds $\sigma_1(\mathbb{R}^2)$.

Step 4: the lower obstruction. One exhibits a single configuration X_* for which no move has positive score at $s = s_*$ and every move has negative score below it. There are 49 possible supports;

for each, the choice of radii is a linear program, so a feasible dual for each support certifies the claim exactly. Five supports tie at zero; the rest are bounded away. Hence $\theta_6 \geq s_*$.

Step 5: the converse, in two halves. For the upper bound one must show that *every* configuration admits a nonnegative move at s_*. This is where the work lies, and it splits cleanly:

- *A failure tree* (Section 7). Only nine of the 49 supports are used. Assuming all nine fail, a sequence of elementary minimax computations — each of which either terminates the branch with a strict, certified separator, or pins down which diameter term is active — reduces the entire hypothesis to three scalar inequalities $q_1, q_2, q_3 \geq 0$ in the four child positions.
- *A weighted geometric lemma* (Section 8). There are exact positive algebraic weights λ, μ for which $\mathcal{W} := q_1 + \lambda q_2 + \mu q_3 \leq 0$ always, with equality only at X_*. Since the failure tree forces $\mathcal{W} \geq 0$, all nine moves cannot fail; and in the boundary case equality forces the configuration to be X_*, where five moves have score exactly zero.

The weighted lemma is itself proved in two pieces: a *self* inequality $\mathcal{W}(X, X) \leq 0$ for a single sibling pair, and a *mixed contraction* $9\mathcal{W}(P, W) \leq 4\{\mathcal{W}(P, P) + \mathcal{W}(W, W)\}$ which reduces the two-pair statement to the one-pair statement. These are the two inequalities carried by interval certificates.

3 Geometric intuition

This section carries no proofs. Its purpose is to explain why the objects above are the right ones and why the answer is the particular algebraic number it is.

3.1 Why disjoint balls at all

Straightness, $\mu(A) \leq \operatorname{diam} A$, is a statement about *one* set. Its power comes from applying it to a union. If $B_1, \ldots, B_k$ are balls with $\mu(B_i) \geq 2\beta r_i$ — which is what a lower density bound gives at small scales — and if the balls are pairwise disjoint, then additivity and straightness together give

$$2\beta \sum_i r_i \;\leq\; \mu\Big(\bigcup_i B_i\Big) \;\leq\; \operatorname{diam}\Big(\bigcup_i B_i\Big).$$

So a configuration of disjoint balls whose radii sum is large relative to the diameter of their union is a *contradiction certificate*: it says the density hypothesis and straightness cannot both hold. The quantity $\sum_i r_i - D/(2s)$ is precisely the amount of contradiction available, and this is the score.

The two-colour structure enters because the balls need not be disjoint across colours: E_1 and E_2 are different sets, and the mass counted in a red ball and the mass counted in a blue ball are different mass. Only same-colour balls must be kept apart. Allowing red–blue overlap is what makes strong configurations possible; it is the source of all the difficulty and all the gain.

3.2 Six points, and why not fewer

With one centre per colour the problem is trivial and gives nothing. With two, one recovers essentially the two-point estimates that underlie the classical bounds. The first genuinely new information comes from allowing each colour a *sibling pair* in addition to its root: two points known to be far apart from one another (distance at least $2s$) but each close to the root (distance at most 1). Straightness is what supplies them — mass in a ball of radius 1 with density bounded below cannot be concentrated near one point, so it must spread, and spreading in a set of finite length means two points at definite separation.

Six is therefore the smallest count at which the geometry has room to express the competition described next.

3.3 The three pressures

Fix the roots at 0 and $e = (1, 0)$, call 0 and its two children red and e and its two children blue. Three families of moves compete.

- **Four children.** Use only the four children, with each sibling pair's radii adding up to its own side length. This move is strong when the two sibling pairs are long and the cross-colour distances are large. It is defeated by moving the children so that some red child comes close to some blue child.
- **A sibling edge against a full triangle.** Use one colour's two children against all three centres of the other colour. This move likes the opposite triangle to be spread out.
- **A root–child edge against a full triangle.** Use a root and one of its children against all three centres of the other colour. This move likes the child to sit far from its own root, which is exactly what weakens the previous two.

Each family can be defeated on its own. The point is that defeating one strengthens another: pulling the children in to shorten cross distances helps the edge–triangle moves, spreading them out helps the four-child move. The extremal configuration is the balance point at which all three pressures are simultaneously neutral — a nonsmooth optimum where several active constraints meet, not a smooth critical point of one function. That is why the answer is an algebraic number of high degree rather than a round one: it is the solution of a system saying "three different moves score exactly zero at once".

3.4 The extremizer, in words

The optimizer turns out to be a *half-turn* configuration: the blue centres are the images of the red centres under the rotation by π about the midpoint $(1/2, 0)$ of the two roots. Both red children sit in the half-plane $x < 0$, on the far side of their own root from the blue root, at mutual distance exactly $c_* = 2s_*$; the blue children sit symmetrically beyond $x > 1$. Every red–blue horizontal separation is then larger than 1, so the configuration is physically admissible with room to spare, and the constraint that binds is not the separation but the interplay of the three moves.

Half-turn symmetry is not imposed. It is *derived*, and the mechanism that derives it is the mixed contraction of Section 8: the contraction inequality rewards the two colours for being aligned, and the only term able to oppose that reward is a crossed mismatch which the residual weighted score exactly dominates. The maximizer of a strictly concave symmetric function must lie on the diagonal, and on the diagonal the two-pair problem collapses to the one-pair problem.

3.5 Why the certificates look the way they do

After the failure tree, what has to be proved is a single inequality $\mathcal{W} \le 0$ for two chords in the unit disk. Two features force the shape of the proof.

First, the inequality is *sharp*: it holds with equality at X_*, so no strict numerical bound can cover a neighbourhood of that point. The certificate is therefore split. Away from the equality configuration a finite cover by boxes with strictly negative upper bounds suffices. Near it, exact differentiation shows that certain coordinates are monotone, which pushes the maximum to a face; on that face the remaining Hessian is proved negative definite, so the function is strictly concave; symmetry then puts the maximum on the diagonal, where the self inequality finishes the job. Concavity plus symmetry, rather than a numerical margin, is what handles the equality point.

Second, the inequality involves Euclidean norms, which are neither polynomial nor everywhere smooth. Positive-coefficient norms are majorized by the tangent-type bound $|z| \le (|z|^2 + \rho^2)/(2\rho)$ and negative-coefficient norms by a supporting linear functional, both valid pointwise on the box in question. What is left is a rational function of the chart parameters, on which outward-rounded interval arithmetic and a rigorous Taylor bound do the rest. Choosing a bad ρ or a bad splitting direction can only cost time; it cannot make a false statement pass, because every step replaces a box by an exact cover of itself.

4 The six-point problem

Normalize the two roots to

$$0 = (0,0), \qquad e = (1,0).$$

Call 0 and its two children *red*, and e and its two children *blue*.

Definition 4.1 (admissible configuration)**.** For $s \in (1/2, 1)$, a configuration of six labelled centres is *admissible at* s if

- the two roots are at distance 1;
- each child is at distance at most 1 from its own root;
- two children with the same root are at distance at least $2s$.

No lower bound on red–blue distances is imposed. The proof below never uses one, so it proves a stronger statement, on the entire product of the two unit child disks. (In the application such bounds are available, since the two colours come from sets at positive distance; they are simply not needed.)

Definition 4.2 (packing move and score)**.** A *packing move* chooses a nonempty set of centres containing at least one red and at least one blue centre, and assigns to every chosen centre a radius $0 \le r_i \le 1$, subject to same-colour disjointness,

$$r_i + r_j \le |x_i - x_j| \qquad \text{for same-colour } i \ne j.$$

Balls of different colours may overlap. The diameter of the union is

$$D = \max_{i,j}\big(|x_i - x_j| + r_i + r_j\big),$$

where $i = j$ is allowed, and the *score* at parameter s is

$$\mathrm{score}_s = \sum_i r_i - \frac{D}{2s}. \tag{1}$$

For a configuration X let $F_s(X)$ be the supremum of score_s over all moves, and let

$$m_6(s) = \inf\{F_s(X) : X \text{ admissible at } s\}.$$

The set of centres used by a move is its *support*. Labelling the three centres of either colour by the bit masks 1 (root), 2 (first child) and 4 (second child), a support is written as a pair of masks: support 67 is the red sibling pair (mask $6 = 2 + 4$) against the full blue triangle (mask 7). There are $7 \cdot 7 = 49$ supports.

Definition 4.3. $\theta_6 = \inf\{s \in (1/2, 1) : m_6(s) \geq 0\}$.

Two elementary remarks will be used repeatedly. For fixed centres and fixed support, maximizing (1) in the radii is a linear program, since D is a maximum of affine functions and all constraints are affine. And for a fixed move, score_s is nondecreasing in s, because every move containing both colours has $D \geq 1 > 0$.

5 The extremal configuration and the exact number

Put $c_* = 2s_*$. The answer is $\theta_6 = s_*$, where

$$c_* = 1.3866128436518097453813568288074219 02\ldots.$$

The decimal is a guide only. The following two equations are the exact definition.

For variables c, B set

$$D = 4c^2 - 2c - B, \qquad b = \frac{2B - 3c^2 + 2c - 1}{c + 1},$$
$$A = \sqrt{\frac{B^2 - 1}{2}}, \qquad C = \sqrt{\frac{B^2 + D^2}{2} - c^2},$$
$$x = \frac{5 - B^2}{4}, \qquad z = \frac{1 + 4b^2 - D^2}{4},$$
$$k = \frac{1 + b^2 - c^2}{2}.$$

Then (c_*, B_*) is the unique solution in the rational box

$$\frac{13866128436518096}{10^{16}} < c < \frac{13866128436518100}{10^{16}}, \quad \frac{2873744161801659}{10^{15}} < B < \frac{2873744161801662}{10^{15}} \tag{2}$$

of the system

$$A + C = 3cb + c^2 - 1, \tag{3}$$
$$(k - xz)^2 = (1 - x^2)(b^2 - z^2), \tag{4}$$

on the branch $x < 0$, $z < 0$, $k - xz < 0$. Finally

$$s_* = \frac{c_*}{2}. \tag{5}$$

Eliminating the square roots in (3) by squaring and clearing denominators produces two polynomial equations, for which an exact-rational Krawczyk inclusion lies strictly inside (2); the displayed sign conditions undo the Gram squaring. Exact intervals also give

$$A^2 > 0, \quad C^2 > 0, \quad R := 3cb + c^2 - 1 > 0, \quad R^2 - A^2 - C^2 > 0,$$

which undo the radical-balance squaring. Together these prove existence and uniqueness on the stated branch. Elimination yields an irreducible degree-54 polynomial for s_*; the two-equation description is exact and considerably more readable.

Define

$$u = (x, \sqrt{1 - x^2}), \qquad v = (z, -\sqrt{b^2 - z^2}).$$

Numerically,

$$\begin{aligned} u &= (-0.814601376872281599, \ \ 0.580021203748434564), \\ v &= (-0.255008890635167971, \ -0.688659775665666991), \\ |v| &= 0.7343581012849697\ldots. \end{aligned}$$

Equation (4), with its branch sign, says exactly that $|u - v| = c_*$. Take the red children to be u, v and the blue children to be their half-turn images $e - u, e - v$.

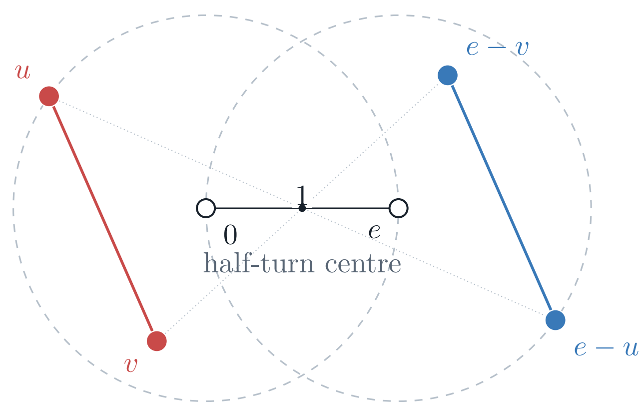

Figure 1: The unique normalized extremizer X_*. The dashed circles are the two unit child disks; each sibling segment has length c_*.

This is a physical configuration. Both x-coordinates of u, v are negative, whereas the blue children have x-coordinate greater than 1. Every non-root red–blue horizontal separation therefore exceeds 1, and the root–child inequalities follow as well.

Four distances reveal why this point is special:

$$A = |e - u|, \qquad B = |e - 2u|, \qquad C = |e - u - v|, \qquad D = |e - 2v|.$$

They obey three balance identities

$$B + D = 2c_*(2c_* - 1), \tag{6}$$

$$A + C = 3c_* b + c_*^2 - 1, \tag{7}$$

$$2B = (c_* + 1)b + 3c_*^2 - 2c_* + 1. \tag{8}$$

These are exactly the zero-score equations for the three mechanisms of Section 3: four children against four children; a sibling edge against a full rooted triangle; and a root–child edge against a full rooted triangle. Half-turn symmetry duplicates the latter two, so five moves tie at zero. Moving the children to defeat the four-child move improves one of the edge–triangle moves, and conversely; the endpoint is the nonsmooth balance point of all three pressures.

6 The obstruction: no constant below s_*

At the configuration X_* of Figure 1, precisely the supports

$$57, \quad 75, \quad 66, \quad 67, \quad 76$$

have score zero; every other support has negative score.

Theorem 6.1 (exact obstruction)**.** *$F_{s_*}(X_*) = 0$, and $F_s(X_*) < 0$ for every $s < s_*$. Consequently $\theta_6 \geq s_*$.*

Proof. The radii of the five displayed moves are explicit, and the three identities (6)–(8) make their scores zero.

For each of the 49 supports the remaining choice of radii is a linear program. One exhibits a feasible dual for one representative of every colour-swap pair. The active duals have value zero. Exact rational interval substitution in (2) makes every inactive dual value smaller than -0.0264. Hence no hidden support has positive score and $F_{s_*}(X_*) = 0$.

If $s < s_*$, the same configuration remains admissible, since its sibling length is $2s_* > 2s$. Every move contains both colours, hence has diameter at least 1, so its score at s is strictly smaller than its score at s_*. Taking the maximum over the 49 supports gives $F_s(X_*) < 0$. □

This is an exact obstruction, not a numerical candidate. What remains is to prove that no other configuration is worse.

7 Nine moves suffice

The converse does not need all 49 supports. We use only

$$\boxed{66,\ 67,\ 76,\ 17,\ 71,\ 37,\ 57,\ 73,\ 75.} \tag{9}$$

Their roles are easier to remember than their masks.

support(s)	geometric role
66	the four children
$67, 76$	a sibling pair against the full opposite rooted triangle
$17, 71$	one root against the full opposite rooted triangle
$37, 57, 73, 75$	a root–child edge against the full opposite triangle

For an arbitrary configuration write the red children as p_1, p_2 and pull the blue children back to the same unit disk by writing their actual locations as $e - w_1, e - w_2$. Thus

$$|p_i| \le 1, \quad |w_j| \le 1, \quad L := |p_1 - p_2| \ge c_*, \quad M := |w_1 - w_2| \ge c_*,$$

and we set

$$r_i = |p_i|, \quad b_i = |w_i|, \quad B_{ij} = |e - p_i - w_j|, \quad a_i = \frac{r_i + b_i}{2}.$$

Here B_{ij} is the actual distance from red child p_i to blue child $e - w_j$.

Theorem 7.1 (nine-packing theorem). *At $s = s_*$, every admissible configuration has a move of nonnegative score among the nine supports (9). Equality is possible only at X_*, up to reflection in the root line, interchange of the colours, and relabelling of the two children within either colour.*

The proof is a failure tree whose purpose is to turn the many possible active diameter constraints into three scalar inequalities.

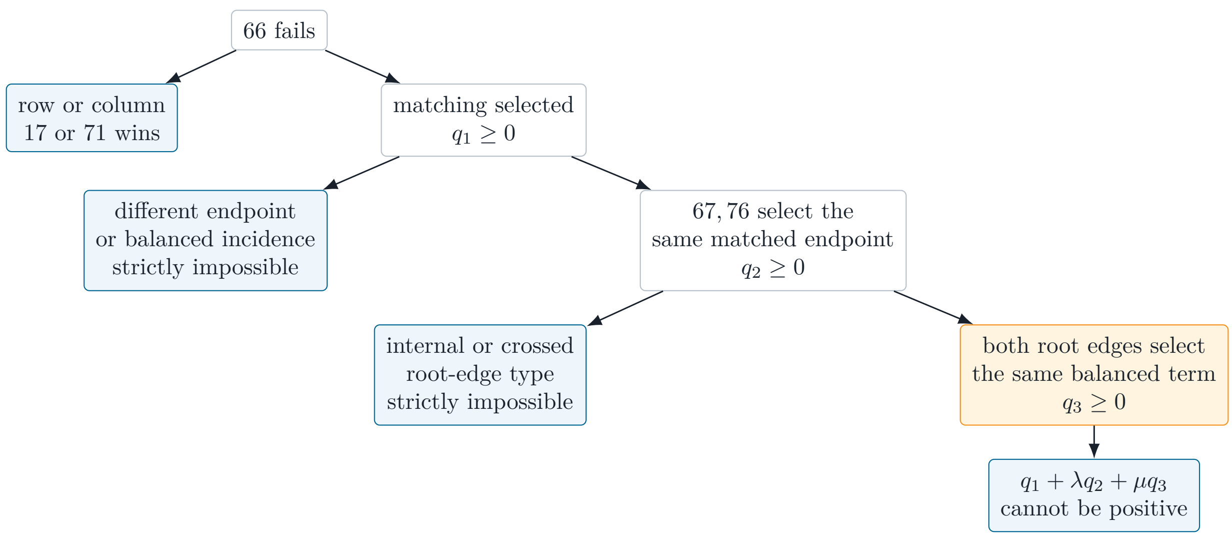

Figure 2: The failure tree. Blue boxes are strict exits. The only surviving path produces the three active slacks of X_*.

7.1 First branch: the four-child minimax

Give each sibling pair radii adding to its actual side length, and write the two radius splits as $x \in [L-1, 1]$, $y \in [M-1, 1]$. Minimizing the four affine red–blue diameter terms over this rectangle gives an elementary alternative: if support 66 has negative score, then at least one of the following holds.

$$B_{i1} + B_{i2} \ge 2 + 2(c_* - 1)L + (2c_* - 1)M, \qquad \text{some row } i, \tag{10}$$
$$B_{1j} + B_{2j} \ge 2 + (2c_* - 1)L + 2(c_* - 1)M, \qquad \text{some column } j, \tag{11}$$
$$B_{11} + B_{22} \ge (2c_* - 1)(L + M), \tag{12}$$
$$B_{12} + B_{21} \ge (2c_* - 1)(L + M). \tag{13}$$

The exact minimax identity is proved in Appendix A. Its other vertices cannot cause failure: a single entry would have to exceed 3, and a same-colour term is short by at least $c_*^2 + 2c_* - 4 > 0$.

A row or column alternative is rescued by support 17 or 71; this is Lemma A.1, proved by a rational tangent bound with margin greater than 0.32. We may therefore relabel the children so that (12) holds, and retain its exact slack

$$q_M := B_{11} + B_{22} - (2c_* - 1)(L + M) \geq 0. \tag{14}$$

Since $L + M \geq 2c_*$, we obtain the slightly weaker but more convenient

$$q_1 := B_{11} + B_{22} - 2c_*(2c_* - 1) \geq q_M \geq 0. \tag{15}$$

7.2 Second branch: the two sibling–triangle moves

For a rooted triangle with child radii r_1, r_2 and side L, use its canonical tangent radii

$$\alpha_0 = \frac{r_1 + r_2 - L}{2}, \qquad \alpha_1 = \frac{r_1 + L - r_2}{2}, \qquad \alpha_2 = \frac{r_2 + L - r_1}{2}, \tag{16}$$

which are nonnegative and at most 1; write β_j for the analogous blue radii.

The one-dimensional radius minimax for support 67 says that a failure witness is either an *endpoint* term or a *balanced* pair of diameter terms; root-labelled and same-colour terms are automatically too short. The same holds for 76.

Fixing the matching in (12) leaves a finite incidence problem. Endpoint/endpoint incidences have six orbits: two coincident, two adjacent, two disjoint. The coincident orbit lying on the selected matching is the desired one. Both disjoint orbits are excluded by a single matching-independent triangle estimate; the two adjacent orbits have separate exact tangent certificates; and the off-matching coincident orbit is strictly excluded by a lens certificate (Section 10).

If at least one witness is balanced there are eight endpoint/balanced and seven balanced/balanced orbits. Eleven are excluded by the two-point rational tangent lemma of Appendix B; the remaining four are strict lens inequalities. Hence both sibling–triangle failures select the same endpoint on the chosen matching; call it B_{11}.

Adding the two exact endpoint-failure inequalities and using $L + M \geq 2c_*$ gives

$$q_2 := B_{11} - \frac{(c_* - 1)a_1 + (c_* + 1)a_2 + 3c_*^2 - 3c_* + 2}{2} \geq 0. \tag{17}$$

The finite incidence check is essential: a balanced witness cannot simply be renamed an endpoint witness.

7.3 Third branch: the two root-edge moves

Keep the two individual endpoint inequalities that produced (17), and consider the root–child edge opposite the selected child together with the full opposite triangle. Its radius minimax initially has sixteen labelled terms; triangle inequalities reduce them pointwise to three types: an internal term, a balanced $(1, 1)$ term, and a balanced $(1, 2)$ term.

The internal term has a rational tangent separator with margin greater than 5. The $(1,2)$ term has a four-variable outward-rounded certificate. Consequently both colour directions must use the $(1,1)$ balanced term, and adding those two failure inequalities yields

$$q_3 := A_1 + C_1 - \big((c_* - 1)a_1 + 3c_* a_2 + c_*^2 - c_*\big) \geq 0, \tag{18}$$

where

$$A_1 = \frac{|e - p_1| + |e - w_1|}{2}, \qquad C_1 = \frac{B_{12} + B_{21}}{2}.$$

Failure of all nine moves therefore forces (15), (17) and (18). At X_* all three are equalities: they are precisely (6)–(8).

8 Why the three failures are incompatible

There are exact positive algebraic numbers

$$\lambda = 0.0894764254088637022746561625951140\ldots,$$
$$\mu = 0.9288383388753901930144096356553482\ldots,$$

defined by the two tangential stationarity equations at (u, v); their short exact definition is Appendix C. Put

$$\mathcal{W}(P, W) = q_1 + \lambda q_2 + \mu q_3. \tag{19}$$

Positivity of the coefficients matters: the failure tree gives $\mathcal{W}(P, W) \geq 0$.

Lemma 8.1 (weighted geometric lemma)**.** *For all ordered sibling pairs $P = (p_1, p_2)$ and $W = (w_1, w_2)$ in the unit disk with sibling lengths at least c_*,*

$$\mathcal{W}(P, W) \leq 0. \tag{20}$$

Equality holds only when $P = W = (u, v)$, or when both pairs are their common reflection in the root line.

The lemma has two parts. The first is a one-pair inequality,

$$\mathcal{W}(X, X) \leq 0, \tag{21}$$

with equality only at (u, v) or its reflection. The second is the mixed contraction: when both sibling lengths equal c_*,

$$9\mathcal{W}(P, W) \leq 4\{\mathcal{W}(P, P) + \mathcal{W}(W, W)\}. \tag{22}$$

Together (21) and (22) give (20) on the chord boundary.

8.1 Reduction to the chord boundary

Expanding (19),

$$\mathcal{W}(P, W) = (1 + \lambda)B_{11} + B_{22} + \mu(A_1 + C_1) - Ua_1 - Va_2 - K, \tag{23}$$

with

$$\begin{aligned} U &= (c_* - 1)\left(\frac{\lambda}{2} + \mu\right), \\ V &= \frac{c_* + 1}{2}\lambda + 3c_*\mu, \\ K &= 2c_*(2c_* - 1) + \frac{\lambda}{2}(3c_*^2 - 3c_* + 2) + \mu(c_*^2 - c_*). \end{aligned}$$

Move p_2 inward along its ray until $|p_1 - p_2| = c_*$. The positive distance terms in (23) change at total speed at most $1 + \mu/2$, whereas the radial penalty changes at speed $V/2$, and exact arithmetic gives

$$\frac{V}{2} - 1 - \frac{\mu}{2} > \frac{1}{2}.$$

So the inward move strictly increases $\mathcal{W}$; the same applies to w_2. Proving the bound after the two moves proves it before them, and equality forces both original sibling lengths to equal c_*.

8.2 What the contraction controls

Put $x_i = e - 2p_i$ and $y_i = e - 2w_i$, and define the midpoint loss and the crossed mismatch

$$\delta(x, y) = \frac{|x| + |y| - |x + y|}{2} \geq 0, \qquad \chi = \frac{|x_1 + y_2| + |y_1 + x_2| - |x_1 + x_2| - |y_1 + y_2|}{4}.$$

A direct cancellation turns (22) into

$$\frac{9\mathcal{W}(P, W) - 4\mathcal{W}(P, P) - 4\mathcal{W}(W, W)}{8} = \frac{\mathcal{W}(P, W)}{8} - (1 + \lambda)\delta(x_1, y_1) - \delta(x_2, y_2) + \mu\chi \leq 0. \quad (24)$$

The two δ terms reward alignment of the two colours. The only term that can oppose them is the crossed mismatch χ, and the small residual $\mathcal{W}/8$ supplies exactly the extra control needed at the extremizer. This is the global mechanism behind the half-turn symmetry.

8.3 The equality case

The global contraction certificate is strict outside a small box in the one chart containing $(P, W) = ((u, v), (u, v))$. In that box the two height variables are strictly increasing, so a maximum lies on the two unit-circle faces. The remaining four-variable Hessian is negative definite, so the contraction expression is strictly concave there. It is symmetric under $P \leftrightarrow W$, hence its unique maximizer lies on the diagonal $P = W$. On that diagonal the left-hand side of (24) equals $\mathcal{W}(P, P)/8$, and the equality statement follows from the self inequality (21).

Proof of Theorem 7.1. Suppose first that all nine selected scores are negative. The failure tree gives $q_1, q_2, q_3 \geq 0$ and therefore $\mathcal{W}(P, W) \geq 0$, while Lemma 8.1 gives $\mathcal{W}(P, W) \leq 0$. Equality is forced throughout, so $P = W = (u, v)$ up to reflection and relabelling. At that configuration five selected moves have score zero, contradicting the assumed strict failure.

For the equality statement, assume instead that every selected score is nonpositive. The row, nonactive-incidence, internal and $(1, 2)$ separators are all strict, so the same failure tree still reaches $q_1, q_2, q_3 \geq 0$, and equality in Lemma 8.1 again forces the displayed half-turn configuration. □

9 Deduction of the optimal constant

Theorem 9.1. *The optimal two-colour six-point constant is*

$$\boxed{\theta_6 = s_* = 0.69330642182590487269067841440371095 1\dots}$$

and the normalized equality configuration is the half-turn configuration of Figure 1, up to its evident symmetries.

Proof. Theorem 6.1 gives $\theta_6 \geq s_*$; it remains to show strict positivity above s_*.

At the endpoint, Theorem 7.1 gives $m_6(s_*) \geq 0$, while the obstruction gives the reverse inequality. Hence $m_6(s_*) = 0$, and the equality statement identifies exactly where this occurs.

Fix $s > s_*$. Its admissible configuration space is compact and is a closed subset of the space admissible at s_*. The best score among the nine moves is continuous in the six centres, being the maximum of finitely many compact linear-program optima. By Theorem 7.1 it is nonnegative at s_*, and its only zeros have sibling length $2s_*$; such configurations are not admissible at s, where the sibling length must be at least $2s > 2s_*$. Compactness therefore gives a uniform positive minimum on the s-admissible space.

Finally the score of every fixed move increases with s, because the coefficient of its positive diameter in (1) decreases. Hence the best nine-move score at parameter s is at least its already positive value computed at s_*, so $m_6(s) > 0$ for every $s > s_*$ and $\theta_6 \leq s_*$. □

10 The finite certificates

This section states exactly where a computer is used. No optimizer, random sample, or floating-point sign is part of the proof. Each computer-assisted claim is a universal inequality on an explicitly covered compact domain.

10.1 The common chord charts

After the radial reductions a sibling pair has exact length c_*. In a local frame it can be written

$$x_1 = a\,n + h\,n^{\perp}, \qquad x_2 = (a - c_*)\,n + h\,n^{\perp}, \tag{25}$$

and the unit-disk condition is exactly

$$c_* - 1 \leq a \leq 1, \qquad |h| \leq \sqrt{1 - \max\{a^2, (a - c_*)^2\}}.$$

Two charts remove the maximum and all trigonometric functions. Put $d = 1 - c_*/2$ and take $t, q \in [0, 1]$. On the two halves of the lens use

$$a = 1 - dt^2 \quad \text{or} \quad a = c_* - 1 + dt^2, \qquad H = t\sqrt{2d - d^2t^2}, \qquad h = (2q - 1)H.$$

For the direction use the rational unit-circle chart

$$n = \left(\varepsilon\frac{1 - z^2}{1 + z^2}, \; \frac{2z}{1 + z^2}\right), \qquad \varepsilon \in \{-1, 1\}. \tag{26}$$

Global reflection lets the first pair use $z \in [0, 1]$; the second uses $z \in [-1, 1]$. The two lens halves and the two signs for each pair give 16 six-dimensional boxes, and (25)–(26) cover the entire chord domain, including overlapping boundaries.

10.2 Outward arithmetic and the covering argument

At precision N an interval is stored as $[m/2^N,\, n/2^N]$ with $m, n \in \mathbb{Z}$. Addition is exact; products and quotients use integer floor and ceiling; square-root endpoints use integer square root. Every operation is therefore rounded outward. The algebraic constants c_*, λ, μ enter only through rational intervals already proved to contain them.

On each box the verifier evaluates a pointwise upper function — either the natural interval extension or a norm-tangent majorant — together with an automatic-differentiation mean-value bound. On smaller unresolved boxes it may also use the rigorous Taylor estimate

$$f(x) \le f(m) + \sum_i |f_i(m)| h_i + \frac{1}{2} \sum_{i,j} \sup_B |f_{ij}|\, h_i h_j, \tag{27}$$

where m is the box midpoint and h_i its half-width. If an upper bound is negative, that box is finished. If a derivative has a fixed sign, the maximum moves to the corresponding face. Otherwise the box is bisected. Every step replaces a box by an exact cover of itself, so a finite list of negative leaves proves the inequality on the whole domain; the split choice affects running time but not validity. If an optional second-derivative expression overflows the integer range, or a rounded derivative denominator meets zero, that Taylor estimate is discarded and the already valid first-order bound is subdivided — such a box is never accepted as a leaf.

For the contraction, positive norms also use

$$|v| \le \frac{|v|^2 + \rho^2}{2\rho},$$

and negative norms use a supporting linear functional. The dyadic ρ and the supporting direction are chosen at the centre of the current box; both inequalities are valid at every point of the box.

10.3 Strict routing inequalities

The routing proof uses the following exact separators. In every row the displayed linear combination has nonnegative coefficients on the failure slacks; except for a zero q_M coefficient in four tangent rows all are positive, and in those four rows the other two coefficients are positive. A strict negative upper bound therefore always rules out simultaneous failure.

branch	proof method	certified margin
four-child row/column	rational two-point tangents	> 0.321
endpoint, disjoint	triangle inequality	> 0.14
endpoint, adjacent $(0,1)$	rational two-point tangents	> 0.00157
endpoint, adjacent $(0,2)$	rational two-point tangents	> 0.00585
11 sibling-incidence cells	rational two-point tangents	> 0.17
4 sibling-incidence cells	exact six-variable lens cover	$> 4.14 \cdot 10^{-8}$
off-matching coincident endpoint	exact six-variable lens cover	$> 1.18 \cdot 10^{-5}$
root-edge internal	rational two-point tangents	> 5
root-edge $(1,2)$	exact four-variable lens cover	$> 6.47 \cdot 10^{-5}$

The structurally delicate strict sibling case is the endpoint/balanced orbit $(E0, S0)$. Its positive separator is

$$F = 17B_{11} + 5B_{22} + 3B_{12} - 3\{(c_* - 1)r_1 + (c_* + 1)r_2\} \\ - \frac{9}{2}\{(c_* - 1)b_1 + (c_* + 1)b_2\} - 9 + \frac{59}{2}c_* - \frac{85}{2}c_*^2. \tag{28}$$

The radial estimates first put both sibling pairs on their chord floors; there, where $L = M = c_*$, the positive combination $5q_M + 9q_E + 6q_S$ simplifies to the displayed F, where q_E is the exact red endpoint slack and q_S the colour-reversed balanced slack whose two endpoints both target red child 1. The exact lens cover proves $F < 0$. The other four compiled cases are handled in the same way.

The frozen 40-bit replay covers 16 charts for each of the five compiled cases, hence 80 charts. It processes 13,269,127 boxes; 6,492,866 leaves are certified negative, and every other processed box is either moved to a monotone face or split into an exact cover. The run makes 751,477 optional Hessian attempts, of which 260 under-resolved attempts are discarded and fall back to subdivision, certifying no leaf. The closest negative upper endpoint over all five cases is $-4.1431121644563973 \cdot 10^{-8}$.

The root-edge $(1, 2)$ separator first uses scalar norm tangents to remove both rotation angles analytically, leaving four shape variables. Its 180-bit replay has 182,403 processed boxes and largest negative upper endpoint $-6.479780941431465 \cdot 10^{-5}$.

10.4 The self inequality

For $X = (p_1, p_2)$, expansion gives

$$\mathcal{W}(X, X) = (1 + \lambda)|e - 2p_1| + |e - 2p_2| \\ + \mu\{|e - p_1| + |e - p_1 - p_2|\} - U|p_1| - V|p_2| - K.$$

Because $V - (2 + \mu) > 1.0417$, moving p_2 inward strictly raises this quantity until $|p_1 - p_2| = c_*$. On that boundary write $r = |p_1|$, $b = |p_2|$, $t = (p_1)_x/r$. The feasible radii are parameterized by a square,

$$r = c_* - 1 + (2 - c_*)u, \qquad b = (c_* - r)(1 - v) + v, \qquad 0 \le u, v \le 1,$$

and the chord equation gives the two possible horizontal coordinates of p_2, the smaller of which always gives the larger value of $\mathcal{W}$. This turns the assertion into one explicit function on $[0, 1]^2 \times [-1, 1]$.

The 180-bit global cover has 169,285 strict leaves and one local box, with largest certified negative upper endpoint $-2.2320712193774615 \cdot 10^{-8}$. On the local box, exact differentiation first puts $|p_1| = 1$; a cover by 201 rectangles then proves

$$\mathcal{W}_{vv} < -0.0870, \qquad \mathcal{W}_{tt} < -3.5406, \qquad \det D^2\mathcal{W} > 0.00400.$$

Hence the remaining two-variable function is strictly concave. The algebraic point (u, v) has value and gradient zero, so it is the unique maximum. This proves (21) and its equality statement.

10.5 The mixed contraction

The identity (24) is evaluated on the same 16 lens charts. After expansion it has six norms with positive coefficients and ten with negative coefficients; on each box the verifier replaces a positive norm by $|z| \le (|z|^2 + \rho^2)/(2\rho)$ and a negative norm by a supporting linear functional, both valid pointwise on the box. The resulting smooth majorant is then bounded by the mean-value and Taylor rules above.

Fifteen charts are strictly negative. The equality chart leaves one rational local box, where exact first derivatives put both height parameters on their top faces, an exact subdivision proves the four-variable Hessian negative definite, and the symmetry-and-diagonal argument following (24) finishes using the self lemma.

For a reproducible replay the equality chart is cut at four exact dyadic midplanes into 16 closed prefix boxes; a separate integer check proves their union is the entire chart, and the boxes overlap only on shared midpoint faces. The 15 ordinary charts and these 16 prefixes report

processed boxes	303,467,033
strict negative leaves	151,733,531
local equality leaves	1
second-order bound attempts	15,536,510

The closest accepted strict upper endpoint is exactly one negative fixed-point unit,

$$-2^{-40} = -9.0949470177292824 \cdot 10^{-13} < 0.$$

On the remaining local leaf the derivative lower margin is 0.0124096, and the negative-definiteness proof has 32,768 interval-LDL leaves with least pivot margin 0.0328339.

Every reported decimal margin is a readable rendering of an exact rational or dyadic endpoint. All sign decisions use the corresponding integer comparisons.

11 From the six-point constant to the density threshold

We record the two steps that turn Theorem 9.1 into Theorem 1.2. Both are formalized in full (Section 12).

Definition 11.1 (straight measure)**.** A Borel measure μ on $\mathbb{R}^2$ is *straight* if $\mu(A) \le \operatorname{diam} A$ for every measurable A.

Definition 11.2 (Besicovitch pair condition)**.** BPC(β) holds if for every straight measure μ there is $\tau > 0$ such that for every scale $\sigma > 0$ there is $\delta > 0$ with the following property. Whenever E_1, E_2 are nonempty measurable sets with $0 < \operatorname{dist}(E_1, E_2) < \delta$ and

$$\mu(B(x,r)) > 2\beta r \qquad \text{for all } x \in E_1 \cup E_2,\ 0 < r < \sigma,$$

there is an open set V meeting both E_1 and E_2 with

$$\mu(V \setminus (E_1 \cup E_2)) > \tau \operatorname{diam} V.$$

Proposition 11.3 (six-point transfer). *Let $0 < s < 1$ and suppose every configuration admissible at s admits a move of nonnegative score, i.e. $m_6(s) \geq 0$. Then* $\mathrm{BPC}(\beta)$ *holds for every $\beta > s$.*

The proof extracts two children near each of two approximate closest points of E_1 and E_2, normalizes the roots to distance 1, applies the six-point property, and absorbs the normalization error using the strict inequality $\beta > s$. Straightness turns the mass in each root ball into a sibling pair of large separation; the physical packing then contradicts straightness by inclusion–exclusion unless the required leaking open set V exists. Only six centres are used.

Proposition 11.4 (pair condition to rectifiability). *If* $\mathrm{BPC}(\beta)$ *holds with $0 < \beta < 1$, then every Borel set of finite $\mathcal{H}^1$ measure whose lower density is at least $\gamma > \beta$ almost everywhere is countably* 1*-rectifiable.*

This is the classical route: decompose into rectifiable and purely unrectifiable parts, localize at a density point of the purely unrectifiable part to obtain a straight measure, apply Definition 11.2 to a maximal disjoint selection, and use continuum surgery together with the fact that a continuum of finite $\mathcal{H}^1$ measure is rectifiable to produce a rectifiable subset of positive measure — contradicting pure unrectifiability.

Proof of Theorem 1.2. Let $c > s_*$ and choose β with $s_* < \beta < c$. By Theorem 9.1, $m_6(s_*) \geq 0$, so Proposition 11.3 gives $\mathrm{BPC}(\beta)$, and Proposition 11.4 shows that c forces rectifiability. Since $c > s_*$ was arbitrary, $\sigma_1(\mathbb{R}^2) \leq s_*$. □

Remark 11.5. The transfer needs only $\beta > s_*$, never the endpoint assertion $\mathrm{BPC}(s_*)$ itself; the infimum in the definition of σ_1 supplies the non-strict conclusion. This is why the endpoint case of the six-point problem — the hardest part of the paper — has to be settled exactly, while the pair condition may be used with room to spare.

12 Formalization in Lean 4

The argument is being formalized in Lean 4 with Mathlib, pinned to `v4.32.0`, in the repository accompanying this paper. The target statements are

`Bescovitch.sigma_one_plane_le_s_star` and `Bescovitch.sStar_le_6934_div_10000`,

namely $\sigma_1(\mathbb{R}^2) \leq s_*$ and $s_* \leq 0.6934$. All definitions occurring in these statements — lower density, countable 1-rectifiability, the threshold σ_1, and the endpoint s_* as the exact isolated radical system of Section 5 — are transparent and are compared recursively against an independent copy, so neither the statement of the theorem nor the meaning of the constant can drift during the proof. The verification gates exclude `native_decide`, `Lean.ofReduceBool`, and custom axioms; the only axioms permitted in the finished theorem are `propext`, `Quot.sound`, and `Classical.choice`. Hashes, program exit codes, and floating-point samples are not proof objects: Bernstein conversion and outward-rounded interval arithmetic are themselves implemented and proved sound inside Lean.

At the time of writing, the entire transfer chain of Section 11 is complete and free of `sorry`, including the geometric-measure-theory results absent from Mathlib (straight decomposition, continuum surgery, finite-length continuum rectifiability, density localization, uniform density on compacta), which are proved locally rather than assumed. The numerical bound $s_* \le 0.6934$ is likewise discharged. The remaining work is confined to the endpoint inequality of Section 8 — in the formalization, the coordinate-free statement `WeightedGeometricBound` and its chart form `WeightedLensChartBound`. Until that hole is closed the Lean development must not be read as a completed proof; its current state is tracked in the repository's development plan.

13 Concluding remarks

Two features of this result deserve emphasis.

First, the constant is *optimal for its instance*. Theorem 1.1 is not an estimate that further effort on the same six-point problem could improve: θ_6 is exactly s_*, and the extremal configuration is unique. Any further progress toward 1/2 along this route must come from a genuinely larger finite problem — more centres, or a different grouping — and not from sharpening the present analysis.

Second, the extremizer is informative. It is a half-turn pair of chords of length c_* pressed against the boundaries of their unit disks, at which three structurally different packing mechanisms tie simultaneously. Any candidate finite problem that hopes to reach below s_* must possess moves that break at least one of those three ties, and the balance identities (6)–(8) say precisely which ones.

A The routing algebra

A.1 The exact two-by-two minimax

Let two same-colour pairs have centre distances $L, M \in [c_*, 2]$. At a score maximizer their two radius sums may be taken to be L and M: raising an unsaturated radius by ε raises the diameter by at most ε, hence raises the score because $1 - 1/c_* > 0$; and the same-colour two-ball primitive dominates either self-ball primitive because $L, M > 1$, so no hidden $2r_i$ term invalidates the increase.

Put $Q = L + M$ and write the red radii as $x, L - x$ and the blue radii as $y, M - y$. The cap $r_i \le 1$ gives $L-1 \le x \le 1$ and $M-1 \le y \le 1$. Minimizing the maximum of the four cross-diameter terms over this rectangle and adjoining the same-colour terms gives exactly

$$\begin{aligned}\Delta_{66} = \max\Big\{&2L,\ 2M,\ Q + B_{ij} - 2,\\ &\frac{Q + L - 2 + B_{i1} + B_{i2}}{2},\ \frac{Q + M - 2 + B_{1j} + B_{2j}}{2},\\ &\frac{Q + B_{11} + B_{22}}{2},\ \frac{Q + B_{12} + B_{21}}{2}\ :\ i, j = 1, 2\Big\}.\end{aligned} \tag{29}$$

A quick proof is linear-programming duality: after centring the two split intervals, a dual distribution on the four entries of the B-matrix has vertices of four kinds — one entry, two entries in a row, two in a column, or two in a perfect matching — and these are exactly the terms in (29). Equating a failed score to $\Delta_{66} \ge c_* Q$ gives (10)–(13).

The two omitted kinds are automatic. In the worse ordering, $c_* Q - 2L \ge c_*(c_* + 2) - 4 > 0$. A single-entry offender would require $B_{ij} \ge 2 + (c_* - 1)Q \ge 2 + 2c_*(c_* - 1) > 3$, whereas $B_{ij} \le 1 + r_i + b_j \le 3$.

A.2 The row/column rescue

Lemma A.1. *If* (10) *holds, support* 17 *has nonnegative score. If* (11) *holds, support* 71 *has nonnegative score.*

Proof. We prove the row statement; colour interchange gives the other. If support 17 also failed, its only possible obstructing primitive would select one of the two blue children; relabel it as child one. We would then have

$$|e-p-w_1|+|e-p-w_2| \geq 2+2(c_*-1)L+(2c_*-1)M,$$
$$|e-w_1| \geq c_*-1+\frac{(c_*-1)(b_1+M)+(c_*+1)b_2}{2}.$$

The root–root and same-colour primitives are strictly below the target, and the root–child assertion follows from the canonical triangle radii (16). Let q_R, q_E denote left side minus right side in the two displayed inequalities; simultaneous failure makes both slacks nonnegative.

Move w_2 inward along its ray. On the region $M \geq c_*$,

$$\frac{dM}{dt} \geq d_0 := \frac{\sqrt{c_*^2-1}}{c_*}, \qquad (2c_*-1)d_0 > 1,$$

so both failure slacks increase until $M = c_*$. The first inequality may also be weakened to $L = c_*$, and convexity puts $|p| = 1$.

For these reduced inequalities the positive combination $10q_R + 9q_E$ is bounded by applying $N \leq (N^2+\rho^2)/(2\rho)$ three times, followed twice by $2|z| \leq \sigma + |z|^2/\sigma$, with rational tangent parameters

$$\rho_1 = \frac{361}{125}, \quad \rho_2 = \frac{64}{25}, \quad \rho_3 = \frac{1937}{1000}, \quad \sigma = \frac{601}{100}, \quad \eta = \frac{567}{100}.$$

The resulting function is convex in (b_1, b_2), so only $(1,1)$, $(1, c_*-1)$, $(c_*-1, 1)$ need be checked; exact rational substitution gives upper bounds below -0.3228344, -0.3213815 and -9.9505 respectively. In particular $10q_R + 9q_E < -8/25$, contradicting simultaneous failure. □

A.3 The sibling and root-edge primitives

The exact endpoint failure slacks used in the second branch are

$$q_{ER}(i,j) = L-1+B_{ij}+\beta_j - c_*(L+T_B),$$
$$q_{EB}(i,j) = M-1+B_{ij}+\alpha_i - c_*(M+T_P),$$

where

$$T_P = \frac{r_1+r_2+L}{2}, \qquad T_B = \frac{b_1+b_2+M}{2}.$$

A balanced red-sibling witness selecting blue children j,k has slack

$$q_{SR}(j,k) = \frac{L+B_{1j}+\beta_j+B_{2k}+\beta_k}{2} - c_*(L+T_B),$$

and the colour-reversed slack is

$$q_{SB}(i,k) = \frac{M+B_{i1}+\alpha_i+B_{k2}+\alpha_k}{2} - c_*(M+T_P).$$

A balanced witness using two different target children already gives one of the two matching inequalities. An endpoint cannot target the triangle root, since

$$c_*(L+T_B) - \{L - 1 + |e - p_i| + \beta_0\} \geq (c_* - 1)L + \left(c_* + \tfrac{1}{2}\right)M - 2 \geq 2c_*^2 - \tfrac{1}{2}c_* - 2 > 1.$$

A balanced root or root–child target is also impossible: its elementary upper bound is at most 6, while the necessary lower bound is $4c_*^2 - c_* > 6$. These observations leave exactly the incidence orbits treated in Section 7.

B The rational two-point tangent lemma

Let x_1, x_2 lie in the unit disk, put $t_i = |x_i|$ and $N_i = |e - 2x_i|$, and assume $|x_1 - x_2| \geq c_*$. For $A_i, \rho_i, \sigma > 0$ and $d_i \geq 0$ put $g_i = A_i/\rho_i$. Then

$$A_1N_1 + A_2N_2 - d_1t_1 - d_2t_2 \leq \max_{(u,v)\in V} \Phi(u,v), \tag{30}$$

where $V = \{(1,1), (1, c_* - 1), (c_* - 1, 1)\}$ and, with $s_1 = u$, $s_2 = v$,

$$\Phi(u,v) = \sum_{i=1}^{2} \frac{A_i}{2\rho_i}(1 + \rho_i^2 + 4s_i^2) + \sigma + \frac{Q}{\sigma} - d_1u - d_2v,$$
$$Q = (g_1 + g_2)(g_1u^2 + g_2v^2) - g_1g_2c_*^2.$$

Indeed, tangent each N_i and observe

$$|g_1x_1 + g_2x_2|^2 = (g_1 + g_2)(g_1t_1^2 + g_2t_2^2) - g_1g_2|x_1 - x_2|^2 \leq Q,$$

then use $2\sqrt{Q} \leq \sigma + Q/\sigma$. The resulting upper bound is convex on the radial polygon $0 \leq t_i \leq 1$, $t_1 + t_2 \geq c_*$, whose relevant vertices are exactly V.

The other elementary ingredient is midpoint convexity,

$$B_{ij} = \left|\frac{(e - 2p_i) + (e - 2w_j)}{2}\right| \leq \frac{|e - 2p_i| + |e - 2w_j|}{2}, \tag{31}$$

which splits every rational separator into one red and one blue application of (30). All tangent parameters in the accompanying checker are rational; after the three radial evaluations, every sign is a rational polynomial sign over (2).

C The exact weights

Write $u = (x,y)$, $v = (z,t)$ with $y > 0 > t$ and $b = |v|$, and retain the distances A, B, C, D of Section 5. Put

$$\kappa = \frac{1 + b^2 - c_*^2}{2}, \qquad R = \sqrt{b^2 - \kappa^2}, \qquad \rho = \frac{b(1-\kappa)}{R}, \qquad z_b = bx - \rho y,$$

$$D_b = \frac{4b - 2z_b}{D}, \qquad C_b = \frac{2b - z_b}{C}.$$

The weights λ, μ are the unique solution of

$$\begin{pmatrix} 2y/B & y/A + (y+t)/C \\ -(c_* + 1)/2 & C_b - 3c_* \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} -2y/B - 2t/D \\ -D_b \end{pmatrix}. \tag{32}$$

These are the two first-order stationarity equations for $q_1 + \lambda q_2 + \mu q_3$ at (u, v). The determinant is $-0.8363707480\ldots$, bounded away from zero by exact intervals, and the same calculation proves

$$\lambda > 0, \qquad \mu > 0, \qquad V - (2 + \mu) > 1.$$

Thus (32), together with the exact algebraic point, defines the weights without recourse to their decimal expansions.

References

[1] A. S. Besicovitch, *On the fundamental geometrical properties of linearly measurable plane sets of points*, Math. Ann. **98** (1928), 422–464.

[2] A. S. Besicovitch, *On the fundamental geometrical properties of linearly measurable plane sets of points II*, Math. Ann. **115** (1938), 296–329.

[3] D. Preiss and J. Tišer, *On Besicovitch's $\frac{1}{2}$-problem*, J. London Math. Soc. (2) **45** (1992), 279–287.

[4] C. De Lellis, F. Glaudo, A. Massaccesi and D. Vittone, *Besicovitch's* 1/2 *problem and linear programming*, arXiv:2404.17536 (2024).

[5] K. J. Falconer, *The Geometry of Fractal Sets*, Cambridge University Press, 1985.

[6] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge University Press, 1995.

[7] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.